

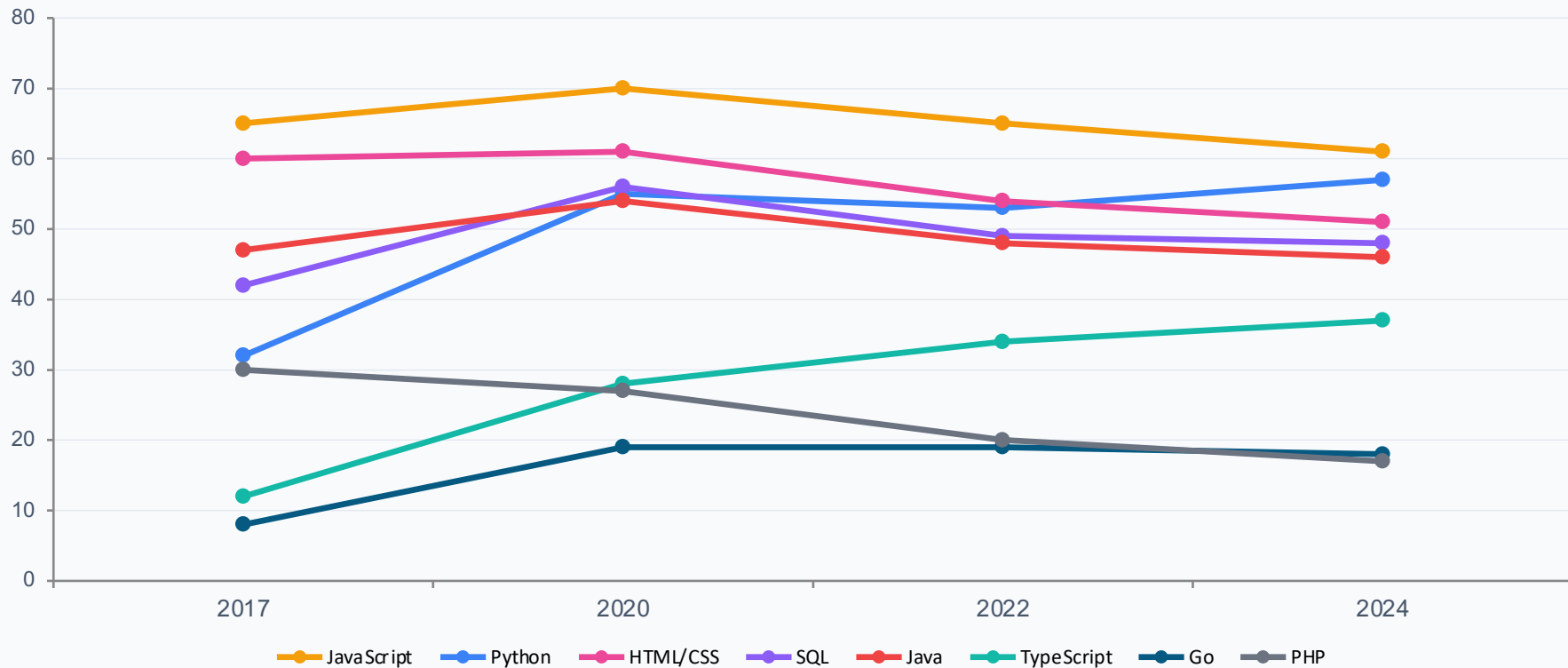
State of Developer Ecosystem 2024:

Language Trends, Satisfaction, and Strategic Shifts

Key insights from JetBrains' annual survey of 23,262 developers worldwide

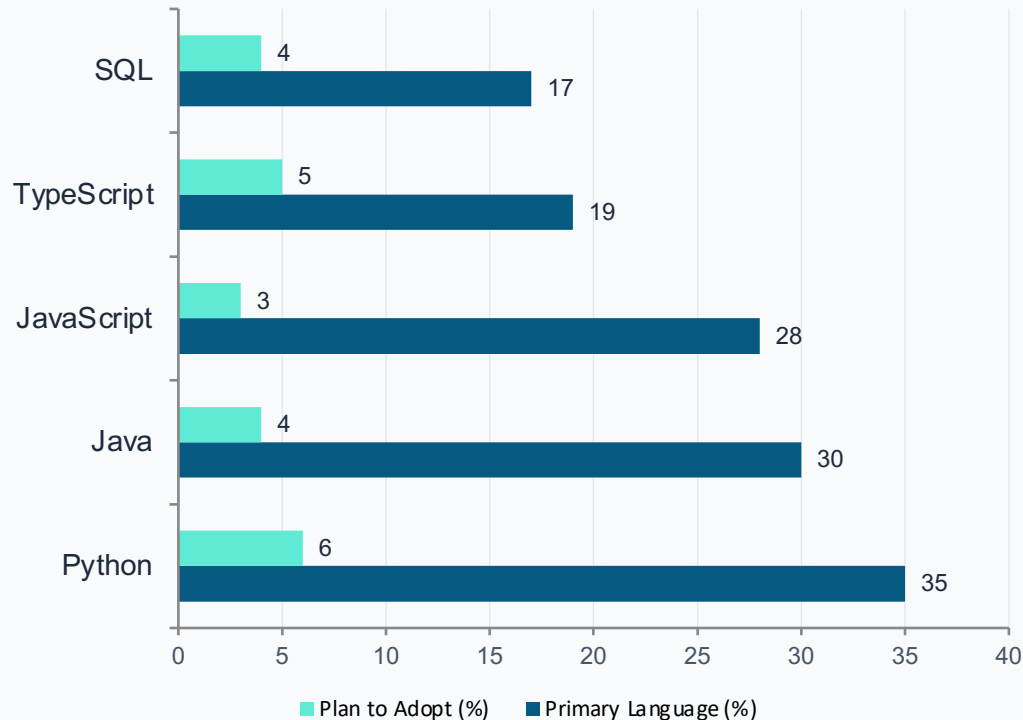
Python Closes the Gap: 8 Years of Language Usage Shifts

Percentage of developers using each language (2017–2024, JetBrains survey)



Python Is Now the #1 Primary Language at 35%

Primary language share vs. adoption intent — developers rarely switch stacks



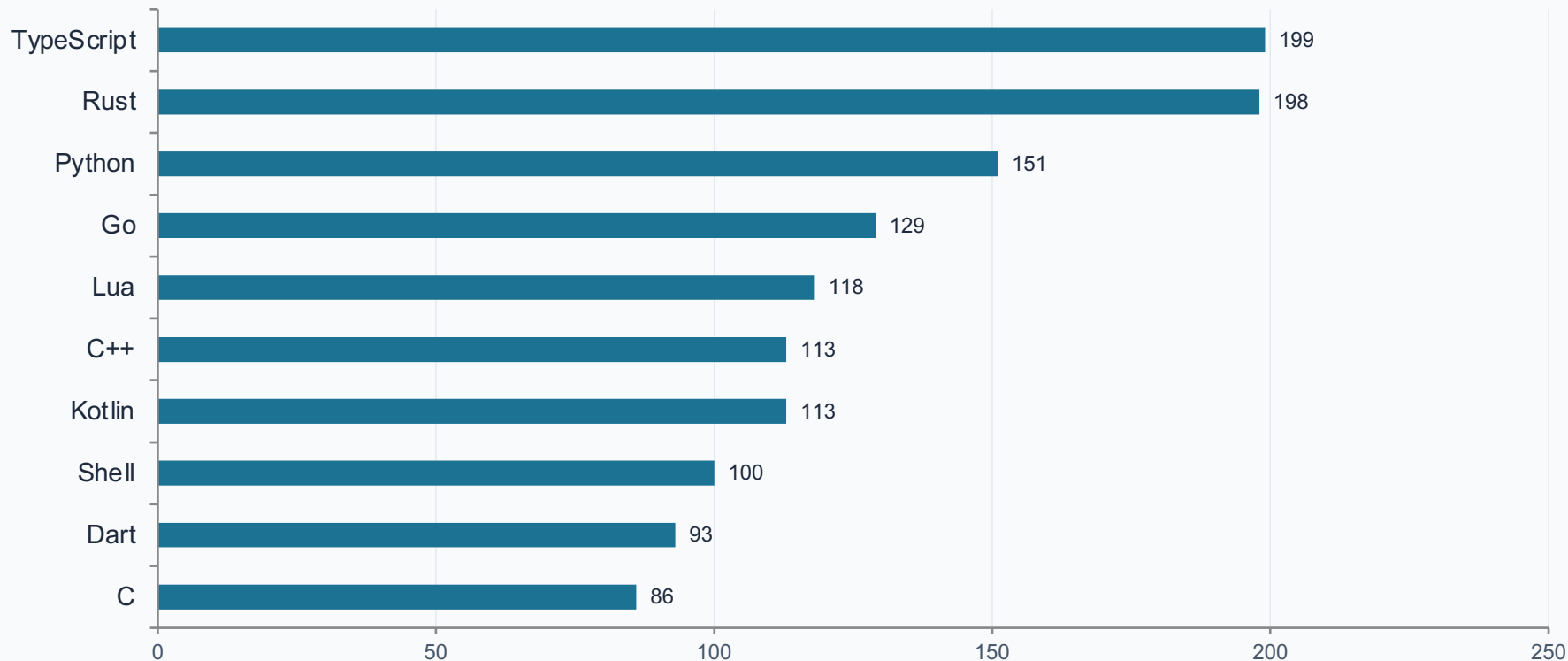
Key Insight

Ecosystem lock-in is real.

Adoption intent remains below 6% across all languages. Developers invest deeply in one stack and rarely plan to switch their primary language.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

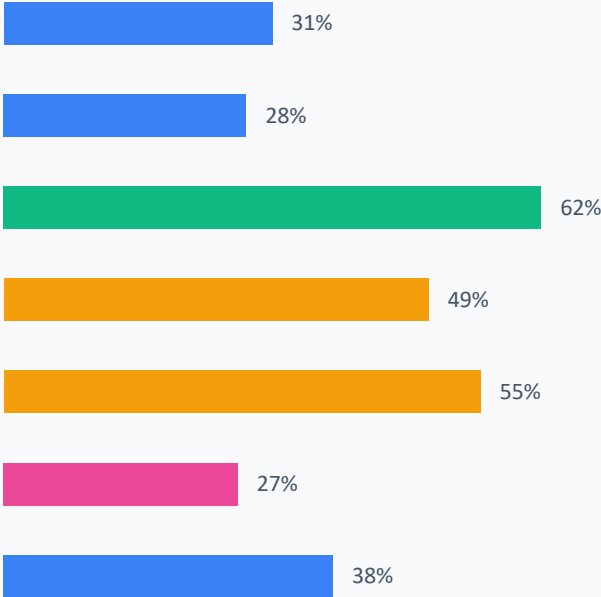
Developer satisfaction scores — type-safe, modern-tooling languages dominate



Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Language choice is driven by application domain, not general preference

Domain	Lead Language	Share
AI / Machine Learning	Python	31%
Data Science & Analytics	Python	28%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic	Java	55%
System Tools & Components	C++	27%
Integrating with APIs	Python	38%



What This Means for Your Technology Strategy in 2025

1

Invest in Python for AI/ML teams

Python is the undisputed leader at 31% for AI and 28% for data science. No alternative comes close.

2

TypeScript is the future of frontend & full-stack

Tripled from 12% to 37% since 2017, leads UI at 62%, and tops developer satisfaction at 199.

3

Rust & Go are growth plays for infrastructure

Both designed for performance and concurrency. Go stable at 18%, Rust rising to 11% — ideal for cloud-native.

4

PHP and Ruby hiring pools are shrinking

PHP dropped from 30% to 17%; Ruby from 10% to 4%. Plan migrations or accept growing talent scarcity.

5

Satisfaction correlates with type safety

Top-2 languages (TypeScript, Rust) both emphasize type safety. Factor modern DX into retention strategy.